

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Revision Test 2026

MATHEMATICS — SSC-I (9th Grade)

Syllabus: Unit 1 - Real Numbers, Unit 2 - Logarithms, Unit 4 - Factorization and Algebraic Manipulation, Unit 5 - Linear Equations and Inequalities, Unit 7 - Coordinate Geometry

Total Marks: 43

Time Allowed: 2 Hours

Version 16

General Instructions:

1. Attempt **ALL** questions. **There are no choices in this paper.**
2. Section A carries 24 marks, Section B carries 9 marks and Section C carries 10 marks.
3. In Section A, fill the correct bubble (A, B, C or D) against each question. Cutting or overwriting a bubble will not be credited.
4. Write all working for Section B and Section C in the answer book. Marks are awarded for correct working, not only for the final answer.
5. Read every part of a question carefully and answer everything it asks for.
6. Use of a calculator is not allowed. Log and antilog tables may be used where required.

SECTION A — Multiple Choice Questions (24 × 1 = 24 Marks)

Attempt **ALL** questions. Fill the correct bubble against each question.

#	Question	A	B	C	D	Answer
1	In $3^5 = 243$, the number 243 is called the:	base	value	exponent	radicand	☐ A ☐ B ☐ C ☐ D
2	$(a^5 \times a^{-3})^2 \div a^{-2}$ simplifies to:	a^2	a^4	a^6	a^8	☐ A ☐ B ☐ C ☐ D
3	On the number line, $\sqrt{19}$ lies between the two consecutive integers:	4 and 5	3 and 4	5 and 6	2 and 3	☐ A ☐ B ☐ C ☐ D
4	The multiplicative identity in the set of real numbers is:	0	-1	any real number a	1	☐ A ☐ B ☐ C ☐ D
5	The Babylonian system of numerals was a sexagesimal system, that is, a system with base:	10	12	60	20	☐ A ☐ B ☐ C ☐ D
6	If $\log x = 2.6180$, then the value of x (found from the antilog table) is:	4.150	415.0	41.50	4150	☐ A ☐ B ☐ C ☐ D
7	$\log_7 7$ is equal to:	1	0	7	undefined	☐ A ☐ B ☐ C ☐ D
8	$\log 15 - \log 3$ is equal to:	$\log 12$	$\log 45$	$\log 18$	$\log 5$	☐ A ☐ B ☐ C ☐ D
9	Written in standard notation, 3.06×10^{-4} is:	0.0306	0.00306	0.000306	0.0000306	☐ A ☐ B ☐ C ☐ D
10	The characteristic of $\log 3456$ is:	4	3	2	0.5386	☐ A ☐ B ☐ C ☐ D
11	The factors of $27x^3 + 64$ are:	$(3x + 4)(9x^2 - 12x + 16)$	$(3x + 4)(9x^2 + 12x + 16)$	$(3x - 4)(9x^2 + 12x + 16)$	$(3x + 4)^3$	☐ A ☐ B ☐ C ☐ D
12	$3x + 3y + ax + ay$ factorizes to:	$3(x + y + a)$	$(3 + a)(x - y)$	$(x + y)(3a)$	$(x + y)(3 + a)$	☐ A ☐ B ☐ C ☐ D
13	$x^3 + 6x^2 + 12x + 8$ factorizes to:	$(x - 2)^3$	$(x + 2)^3$	$(x + 2)(x^2 - 2x + 4)$	$(x + 4)(x + 2)^2$	☐ A ☐ B ☐ C ☐ D
14	$x^2 + 10x + 25 - y^2$ factorizes to:	$(x + 5 + y)(x + 5 - y)$	$(x + 5 + y)(x - 5 - y)$	$(x - 5 + y)(x - 5 - y)$	$(x + y + 5)(x + y - 5)$	☐ A ☐ B ☐ C ☐ D
15	The complete factorization of $81m^4 - 16n^4$ is:	$(9m^2 - 4n^2)(9m^2 + 4n^2)$	$(3m - 2n)^4$	$(9m^2 + 4n^2)(3m + 2n)(3m - 2n)$	$(9m^2 + 4n^2)(3m + 2n)^2$	☐ A ☐ B ☐ C ☐ D
16	The factors of $3x^2 + 10x + 8$ are:	$(3x + 2)(x + 4)$	$(3x + 8)(x + 1)$	$(3x - 4)(x - 2)$	$(3x + 4)(x + 2)$	☐ A ☐ B ☐ C ☐ D
17	Which of the following is NOT a linear equation in one variable?	$2x + 3 = 0$	$x/2 - 1 = 3$	$x^2 = 4$	$5x + 3 = 5$	☐ A ☐ B ☐ C ☐ D
18	If $a < b$, then $a + c < b + c$ for every real number c. This is the:	additive property of inequality	trichotomy property	transitive property	multiplicative property of inequality	☐ A ☐ B ☐ C ☐ D
19	The solution of the inequality $(2x/3) - 1 < 3$ is:	$x < 3$	$x < 6$	$x > 6$	$x < 2$	☐ A ☐ B ☐ C ☐ D

20	If $\sqrt{3x - 2} = 4$, then x is equal to:	2	4	16/3	6	Ⓐ Ⓑ Ⓒ Ⓓ
21	The abscissa of every point lying on the y-axis is:	1	equal to its ordinate	zero	-1	Ⓐ Ⓑ Ⓒ Ⓓ
22	If the ordered pairs $((a + 1)/2, b - 3)$ and $(4, 5)$ are equal, then the value of a is:	4	7	8	9	Ⓐ Ⓑ Ⓒ Ⓓ
23	If B is the mid-point of the line segment AC, then AC : BC is:	1 : 1	1 : 2	3 : 1	2 : 1	Ⓐ Ⓑ Ⓒ Ⓓ
24	The distance between the points $(-6, 3)$ and $(6, 8)$ is:	13 units	12 units	5 units	17 units	Ⓐ Ⓑ Ⓒ Ⓓ

SECTION B — Short Questions ($3 \times 3 = 9$ Marks)

Attempt **ALL** questions. There are no choices.

Part	Question	Marks
2(i)	(a) Simplify, using the laws of exponents: $(x^5 \times x^{-2}) \div x^4$ [1] (b) Evaluate: $(81)^{3/4}$ [1] (c) Write $\sqrt[4]{a^7}$ in exponential form. [1]	3
2(ii)	Given that $\log 2 = 0.3010$ and $\log 3 = 0.4771$, evaluate log 48 . Show every step of your working and name the law of logarithms you use at each step.	3
2(iii)	Solve the equation $ 2z - 3 = 9$ and verify both of your solutions by substituting them back into the original equation.	3

SECTION C — Long Questions ($2 \times 5 = 10$ Marks)

Attempt **ALL** questions. There are no choices.

Part	Question	Marks
3(i)	(a) Factorize completely: $1 + 4x^4$ [2] (b) Find the square root of $49p^2 - 84pq + 36q^2$ by factorization. [1] (c) Simplify: $(x^2 - 64)/(x^2 + 11x + 24) \div (x^2 - 16x + 64)/(x^2 + 3x)$ [2]	5
3(ii)	A(-1, -2), B(-1, 4) and C(7, 4) are the vertices of a triangle. (a) Use the distance formula to find the lengths AB, BC and AC. [2] (b) Show that $\triangle ABC$ is a right triangle and name its hypotenuse. [1] (c) Find the coordinates of the mid-point M of the hypotenuse and show that M is at an equal distance from all three vertices A, B and C. [2]	5

Invigilator's Signature: _____

Candidate's Signature: _____

Units covered: 1 Real Numbers • 2 Logarithms • 4 Factorization and Algebraic Manipulation • 5 Linear Equations and Inequalities • 7 Coordinate Geometry

— END OF QUESTION PAPER —